CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Tech. (CL/EC/EE/ME/IT/CE) Examination

May-2014

MA 101: Engineering Mathematics-I

Date: 15.05.14, Thursday **Time: 01:30 p.m. to 04:30 p.m.** **Total: 70 Marks**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 (4+3)

(a) Check the convergence of $\sum_{n=0}^{\infty} \frac{5^n + 4^n}{6^n}$

(b) Express $f(z)$ in z where $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

Q-2 (4+5+5)

(a) Discuss the convergence of infinite series $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{1+n^2}$

(b) For the value of which value of μ and λ so that the equations:

$2x + y + 2z = 5, \quad x + 2y + 3z = 6, \quad x + 2y + \lambda z = \mu$

have (a) no solution (b) Unique solution (c) infinite number of solutions.

(c) Discuss the convergence of infinite series $\sum_{n=0}^{\infty} \frac{n!\, 2^n}{n^2}$

OR

Q-2 (4+5+5)

(a) Test the convergence of the following series $\frac{1}{2!}+\frac{2}{3!}+\frac{3}{4!}+\cdots+\infty$

(b) Test the consistency of the following system and if consistent then obtain solution by Gauss Elimination method:

$$x+2y+z=5,\quad 3x-y+z=6,\quad x+4y+z=7$$

(c) Using Gauss Jordan Method, find inverse of matrix $A=\begin{bmatrix}8&4&3\\2&1&1\\1&2&1\end{bmatrix}$

Q-3 (4+5+5)

(a) Check the converges of the following series $\sum_{n=0}^{\infty}\left(\frac{n+1}{n+2}\right)^{2}x^{n}$

(b) Write the function $f(z)=z+\frac{1}{z}$ in the form of $f(z)=u(r,\theta)+iv(r,\theta)$

(c) Define the rank of matrix. Find the rank of following matrix by using Minors (determinant) $A=\begin{bmatrix}1&2&3\\1&4&2\\2&6&5\end{bmatrix}$

OR

Q-3 (4+5+5)

(a) Reduce each of these quantities to a real number $(1-i)^4$

(b) Locate the numbers z_1+z_2 and z_1-z_2 in vector form when

i) $z_1=2i$, $z_2=\frac{2}{3}-i$ ii) $z_1=(-\sqrt{3},1)$, $z_2=(\sqrt{3},0)$

(c) If possible then convert following matrix in row-echelon form $A=\begin{bmatrix}0&2&1\\5&2&3\\1&1&0\end{bmatrix}$

SECTION-II

Q-4 (4+3)

(a) Evaluate: $\lim_{x \to 1} \frac{1+\log x - x}{1-2x+x^2}$

(b) Evaluate: $J = \frac{\partial(u,v)}{\partial(x,y)}$ for $u = x^2, \quad v = y^2$

Q-5 (4+5+5)

(a) Evaluate: $\lim_{x \to 0} \frac{x\cos x - \log(1+x)}{x^2}$

(b) Find the n^{th} derivative of $\frac{x}{(x-1)(2x+3)}$

(c) Expand $f(x) = x^3 + 4x^2 + 8x + 1$ in powers of $(x-1)$

OR

Q-5 (4+5+5)

(a) If $y = (2-3x)^{10}$ then find y_9

(b) State and prove Euler's theorem for homogeneous function.

(c) Expand $f(x) = x^4 - 5x^3 + 5x^2 + x + 2$ in powers of $(x-2)$

Q-6 (4+5+5)

(a) Find total differential of $w = 4x^3 - xy^2 + 3y - 7$

(b) Find the equation of the tangent plane and normal line to the surface $x^2 + 2y^2 + 3z^2 = 12$ at $(1,2,-1)$

(c) Find approximate error in its surface of rectangular parallelepiped of sides a, b and c in error δ is made in the measurement of each side.

OR

Q-6 (4+5+5)

(a) Find approximate value of $\sqrt{(299)^2 + (399)^2}$.

(b) If $u = \sin^{-1}\frac{x^2+y^2}{x+y}$ then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$

(c) If the sides and angles of a plane triangle vary in such a way that its circum radius remains constant, prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos c} = 0$